

Quasi-Experimental Designs

Studying possible effects when random assignment is not feasible

WEEK 8 • DAY 2 • QUASI-EXPERIMENTS

Reminder: what made an RCT strong?

- The researcher manipulated a treatment or condition.
- Participants were randomly assigned to treatment or control.
- The groups were compared on an outcome.
- Random assignment helped reduce alternative explanations.

RCTs are useful for causal inference—but they are not always possible.

When random assignment is not possible

- We cannot randomly assign people to experience a real crisis.
- We may not control how a campaign, platform change, or news event unfolds.
- But we may still want to study whether something appears to have an effect.

This is where quasi-experiments become useful.

What is a quasi-experiment?

- A quasi-experiment studies a possible effect WITHOUT random assignment.
- It often uses NATURALLY occurring groups, events, or timing.
- It tries to create a reasonable comparison.
- Because groups are not randomized, alternative explanations are harder to rule out.

Quasi means “almost” or “resembling”—not fully randomized.

RCT vs. quasi-experiment

Feature	RCT	Quasi-experiment
Random assignment?	Yes	No
Treatment / exposure?	Researcher usually controls it	Often naturally occurring
Comparison group?	Usually built into the design	Must be carefully chosen
Causal confidence	Stronger if well designed	More cautious; threats remain

Do NOT call a quasi-experiment an RCT if there is no random assignment.

Examples in communication research

- A university launches an AI-use disclosure policy in classes.
- A brand releases an apology campaign after a crisis.
- A public health campaign runs in one county but not another.
- A platform adds a misinformation warning label for some content.

Many real-world communication changes CAN NOT be randomized by the researcher.

Design 1: one-group pretest-posttest

- Measure one group before the change.
- The change or intervention happens.
- Measure the same group after the change.
- Example: measure student trust in university emails before and after a new crisis communication policy.

Simple, but weak if used alone.

Design 1: One-group pretest-posttest

Example: thread-style campaign

How it works



- 1 Measure one group before the change.
- 2 The campaign or event happens.
- 3 Measure the same group again after.

Communication example



Before campaign

Thread campaign launches

After campaign



Outcome: stigma attitude score

- 1 Measure student stigma attitudes before the campaign.
- 2 Measure the same students again after the campaign.

 **Key idea:** Did the outcome change after the intervention?



Main limitation: something else may have caused the change.

Threat: history

- History means another event happens around the same time as the intervention.
- Example: trust increases after the apology campaign.
- But during the same week, a beloved celebrity praises the brand.
- The increase may not be caused by the apology alone.

History threat = outside events may explain the change.

Threat: maturation or natural trends

- People can change over time even without the intervention.
- Example: students become more comfortable with AI tools over the semester.
- If attitudes improve after a workshop, some change may reflect natural learning over time.
- A pre-post design must consider **what would have happened anyway.**

Ask: would the outcome have changed even without the intervention?

Design 2: Nonequivalent comparison group

Example: thread-style campaign across two campuses

How it works



1

Compare a group exposed to the campaign with a group not exposed.

2

The groups are NOT randomly assigned.

3

Compare their outcome scores after the campaign.



Key question: Are the groups already different before the campaign?

Communication example



Campus A
students

Sees
campaign

Stigma
attitude
score

Campus B
students

No
campaign

Stigma
attitude
score



Outcome: stigma attitude score

1

Measure stigma attitudes at Campus A after the campaign.

2

Measure the same outcome at Campus B, which did not receive the campaign.



Key idea: Compare an exposed group to a similar non-exposed group.



Main limitation: selection bias — the groups may differ even before the campaign.

Nonequivalent comparison group

Group	Exposure	Outcome
Campus A	Sees mental-health campaign	Stigma attitude score
Campus B	Does not see campaign	Stigma attitude score
Question	Is Campus A different from Campus B after the campaign?	Could the campuses already differ?

Because the groups are not randomized, check how similar they are.

Threat: selection bias

- Selection bias means the groups differ before the intervention.
- Example: Campus A volunteers for the campaign because it already prioritizes mental health.
- Campus B does not volunteer and has different student attitudes.
- A later difference may reflect pre-existing group differences.

The comparison group must be justified.

Design 3: Pretest-posttest with comparison group

Example: thread-style campaign across two campuses

How it works



- 1 Measure BOTH groups before the campaign.
- 2 One group receives the campaign; the comparison group does not.
- 3 Measure BOTH groups again after.
- 4 Compare which group changed more.



Key question: Did the campaign group change more than the comparison group?



Key idea: Compare changes over time in an exposed group and a non-exposed group.

Communication example



Outcome: stigma attitude score

- 1 Measure stigma attitudes on both campuses before the campaign.
- 2 After the campaign, measure both campuses again and compare which campus changed more.



Main limitation: the groups may still differ before the campaign, and other events may affect change.

Simple difference-in-differences intuition

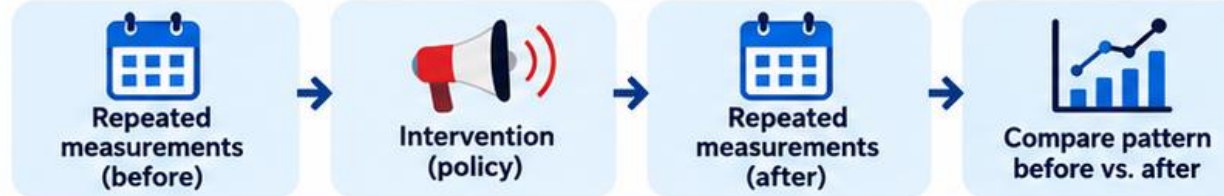
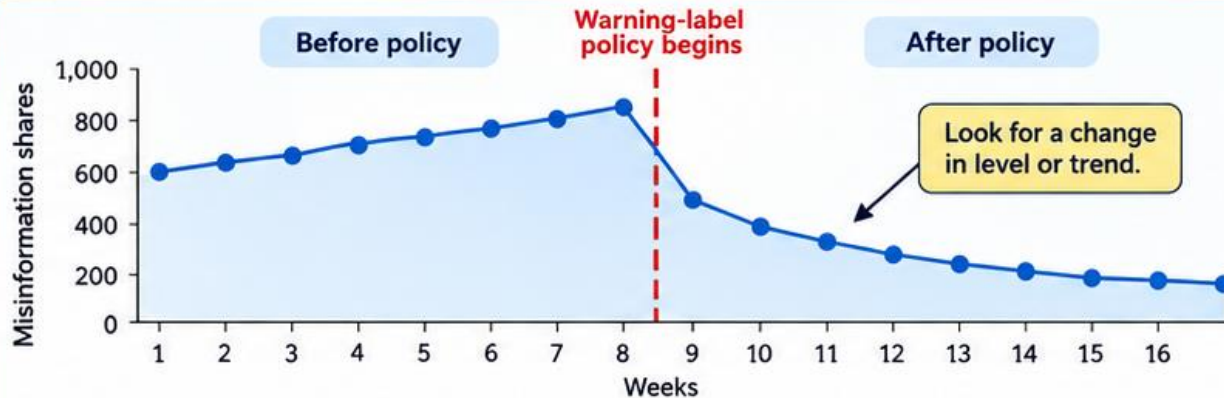
Group	Before	After	Change	Meaning
Campaign campus	50	65	+15	Exposure
Comparison campus	52	56	+4	Trend
Difference	—	—	+11	Estimated effect

The key idea: compare changes, not just final scores.

Design 4: Interrupted time series

Example: platform warning-label policy and weekly misinformation sharing

How it works



1 Measure the same outcome many times before the event.

2 The intervention or policy happens.

3 Keep measuring the same outcome many times after.

4 Look for a change in level or trend.

? **Key question:** Did the pattern change after the intervention point?

Communication example



Weeks 1-8

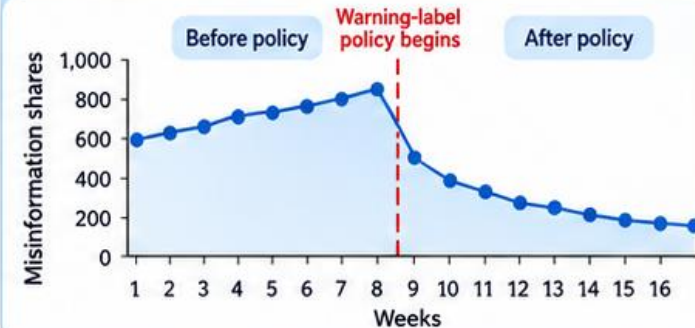
Measure weekly misinformation shares

Platform adds warning labels (week 9)

Weeks 9-16

Continue measuring weekly misinformation shares

Outcome: weekly misinformation shares



1 Track weekly misinformation sharing before the warning-label policy.

2 After labels are added, continue tracking and examine whether the pattern changes.



Key idea: Use many time points to compare the pattern before and after a real-world intervention.



Main limitation: another event at the same time could also explain the change.

Interrupted time series: what to look for

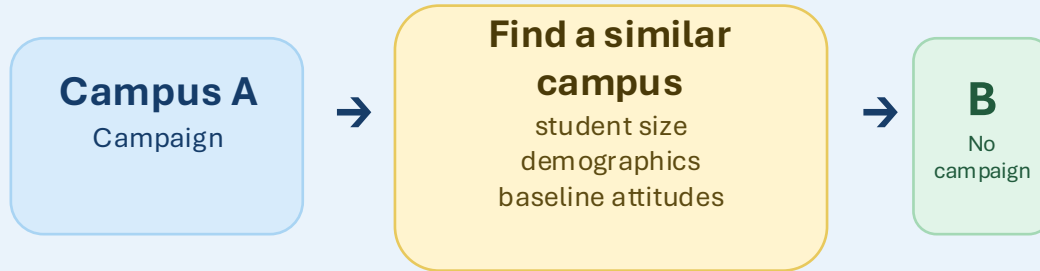
- Before the event: what was the normal trend?
- Immediately after the event: did the level jump or drop?
- Over time after the event: did the trend change?
- Were there other major events at the same time?

Time helps you see whether the change is unusual.

Matching: making the comparison group more reasonable

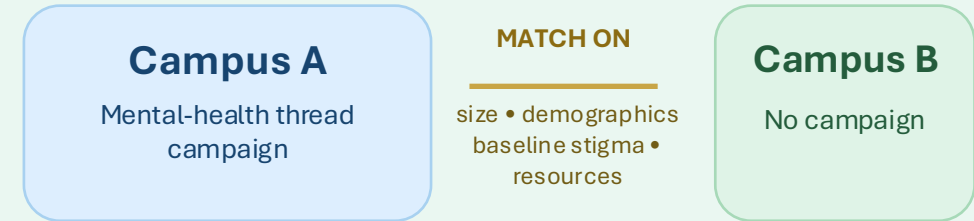
A simple way to reduce some selection bias in quasi-experiments

How matching works



- 1 Start with an exposed group and a potential comparison group.
- 2 Choose a comparison group that is similar on important measured characteristics.
- 3 Then compare outcomes between the groups.

Communication example



Outcome: stigma attitude score after the campaign

Matching can make the comparison more reasonable because the campuses look more alike on measured characteristics.

Important: matching cannot fix differences you did not measure.

Key idea: Matching reduces some selection bias — but it does not create random assignment or make the study an RCT.

Choosing among quasi-experimental designs

- Do you have measurements before and after the event?
- Do you have a reasonable comparison group?
- Do you have multiple time points?
- Can you show groups were similar before the intervention?
- Can you identify major alternative explanations?

The best quasi-design depends on the data you can reasonably collect.

What data / situation do you have?	Design to consider	Basic logic
One group + before and after measurements	One-group pretest-posttest	Did the group change after the intervention?
Exposed group + non-exposed comparison group	Nonequivalent comparison group	Are the two groups different after the intervention?
Two groups + before and after measurements	Pretest-posttest with comparison group	Did the exposed group change more than the comparison group?
Many measurements before and after an event	Interrupted time series	Did the level or trend change after the intervention?
Groups differ, but you can find similar comparison cases	Matching	Can we compare cases that were similar on important characteristics?

Activity: choose the quasi-design

A university introduces a new AI disclosure policy for student newspaper articles in Fall 2026.

1. What outcome could you measure?
2. Could you measure before and after the policy?
3. What would be a reasonable comparison group?
4. What alternative explanations should you mention?
5. Would you call this an RCT? Why or why not?

Causal language: be careful

- Strong RCT: “The assigned message condition **caused** a difference in credibility ratings in this study.”
- Good quasi-experiment: “The policy **was associated** with a change, compared with the comparison group.”
- Weak pre-post study: “Credibility changed after the campaign, **but other explanations remain possible.**”
- Do NOT overclaim beyond the design.

Write the conclusion your design can support.

Validity threats in quasi-experiments

Threat	Meaning	Example
Selection	Groups differ before the intervention	Campaign county already more health-conscious
History	Other event happens at same time	News scandal occurs during campaign
Maturation	People change naturally over time	Students learn more across semester
Instrumentation	Measurement changes over time	Different survey question used after
Attrition	Different people drop out	Less-engaged participants stop responding

Name the threat and explain why it matters for your study.

How to strengthen a quasi-experiment

- Collect baseline data before the intervention when possible.
- Use the same measures before and after.
- Add a comparison group that is as similar as possible.
- Use multiple time points instead of only one before and one after.
- Be transparent about remaining limitations.

Quasi-experiments are about building the best available comparison.

RCT and quasi-experiment: both can be useful

RCT is better when...

- You can ethically and practically assign conditions.
- You can control the message or intervention.
- Your main goal is causal testing.

Quasi is better when...

- The event already happened or cannot be assigned.
- You need to study a real-world policy, platform change, or campaign.
- Randomization would be unethical or impossible.

The right design depends on the question and constraints.

Which design fits?

Research question	Likely design	Why
Does an AI-generated label reduce credibility of the same news post?	RCT	Can randomize post version
Did a campus campaign change attitudes from September to November?	Pre-post quasi	Measures over time
Did County A campaign reduce hotline misinformation more than similar County B?	Comparison quasi	Uses similar non-exposed group
Did platform warning labels change weekly misinformation sharing?	Time series	Uses many time points

Design follows the research question.

Mini quiz

- 1. What is the key feature missing from a quasi-experiment?
- 2. Why is a comparison group useful?
- 3. What is one threat to validity in a one-group pre-post design?
- 4. Why should we avoid strong causal language in weak quasi-experiments?
- 5. When might a quasi-experiment be more realistic than an RCT?

Answer without looking back.

Takeaway

- RCTs use random assignment to create comparable conditions.
- Quasi-experiments study possible effects without random assignment.
- Quasi-experiments need careful comparison, timing, and validity discussion.
- Your Assignment 4 design should be simple, clear, and defensible.

Causal claims are only as strong as the design supporting them.

Pair activity: strengthen this weak design

Weak plan: “We will survey students after a university campaign and ask whether the campaign helped them trust official emails.”

1. What is missing from this design?
2. Could you add a before measure?
3. Could you add a comparison group?
4. What outcome would you measure clearly?
5. What limitation would still remain?